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2O13

OOPS Assignment-1

1. WAP to display "Hello World"

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
using namespace std;

int main() {
    cout << "Hello World";
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd
Hello World
PS C:\Users\Shagun Garg>
```

2. WAP to input an integer, decimal and character and display it

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
using namespace std;

int main() {
    int i;
    float f;
    char c;

    cout << "Enter integer: ";
    cin >> i;
    cout << "Enter decimal: ";
    cin >> f;
    cout << "Enter character: ";
    cin >> c;

    cout << "Integer: " << i << endl;
    cout << "Decimal: " << f << endl;
    cout << "Character: " << c << endl;

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "
Enter integer: 5
Enter decimal: 3.14
Enter character: $
Integer: 5
Decimal: 3.14
Character: $
PS C:\Users\Shagun Garg>
```

3. WAP to add, subtract, multiply and divide numbers

```
ers > Shagun Garg > oops.cpp > ...
#include <iostream>
using namespace std;

int main() {
    float a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;

    cout << "Sum = " << a + b << endl;
    cout << "Difference = " << a - b << endl;
    cout << "Product = " << a * b << endl;
    cout << "Division = " << a / b << endl;

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd
Enter two numbers: 9
4
Sum = 13
Difference = 5
Product = 36
Division = 2.25
PS C:\Users\Shagun Garg>
```

4. WAP to convert Celsius to Fahrenheit

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
using namespace std;

int main() {
    float c, f;
    cout << "Enter temperature in Celsius: ";
    cin >> c;

    f = (9 * c / 5) + 32;
    cout << "Temperature in Fahrenheit = " << f;

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Sha
Enter temperature in Celsius: 36
Temperature in Fahrenheit = 96.8
PS C:\Users\Shagun Garg>
```

5. WAP to format console output using \n, \t, \b, endl, setw.

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
#include <iomanip>
using namespace std;

int main() {
    cout << "Hello\nWorld" << endl;
    cout << "Hello\tWorld" << endl;
    cout << "Helloo\b" << endl;
    cout << setw(10) << "Shagun";

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\
Hello
World
Hello   World
Helloo
    Shagun
PS C:\Users\Shagun Garg>
```

6. WAP to utilize assignment += and -= operators

```
Users > Shagun Garg > oops.cpp > main()
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    a += 5;
    cout << "Addition  : " << a << endl;
    a -= 3;
    cout << "Subtraction : " << a << endl;
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\
Addition  : 15
Subtraction : 12
PS C:\Users\Shagun Garg>
```

7. WAP to swap two numbers using bitwise operator

```
Users > Shagun Garg > oops.cpp > ...
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    a = a ^ b;
    b = a ^ b;
    a = a ^ b;
    cout << "After swapping:\n";
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:
Enter two numbers: 15
10
After swapping:
a = 10
b = 15
PS C:\Users\Shagun Garg>
```

8. Library fine problem.

Users > Shagun Garg > oops.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main() {
    int days;
    cout << "Enter number of days late: ";
    cin >> days;
    if (days <= 5)
        cout << "Fine = " << days * 0.5 ;
    else if (days <= 10)
        cout << "Fine = " << days * 1 ;
    else if (days <= 30)
        cout << "Fine = " << days * 5 ;
    else
        cout << "Membership cancelled";

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\User
Enter number of days late: 31
Membership cancelled
PS C:\Users\Shagun Garg> cd "c:\User
Enter number of days late: 14
Fine = 70
PS C:\Users\Shagun Garg>
```

9. WAP to display arithmetic operations .

Users > Shagun Garg > oops.cpp > main()

```cpp
using namespace std;

int main() {
    int a, b;
    char op;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Enter operator (+ - * /): ";
    cin >> op;
    switch (op) {
        case '+': cout << a + b;
        break;
        case '-': cout << a - b;
        break;
        case '*': cout << a * b;
        break;
        case '/': cout << a / b;
        break;
        default: cout << "Invalid operator";
    }
    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\
Enter two numbers: 8
3
Enter operator (+ - * /): ?
Invalid operator
PS C:\Users\Shagun Garg> cd "c:\
Enter two numbers: 8
3
Enter operator (+ - * /): /
2
PS C:\Users\Shagun Garg>
```

10. WAP to check whether a number is even or odd.

Users > Shagun Garg > oops.cpp > main

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter number: ";
    cin >> n;

    if (n % 2 == 0)
        cout << "Even number";
    else
        cout << "Odd number";

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "
Enter number: 79
Odd number
PS C:\Users\Shagun Garg> cd "
Enter number: 828
```